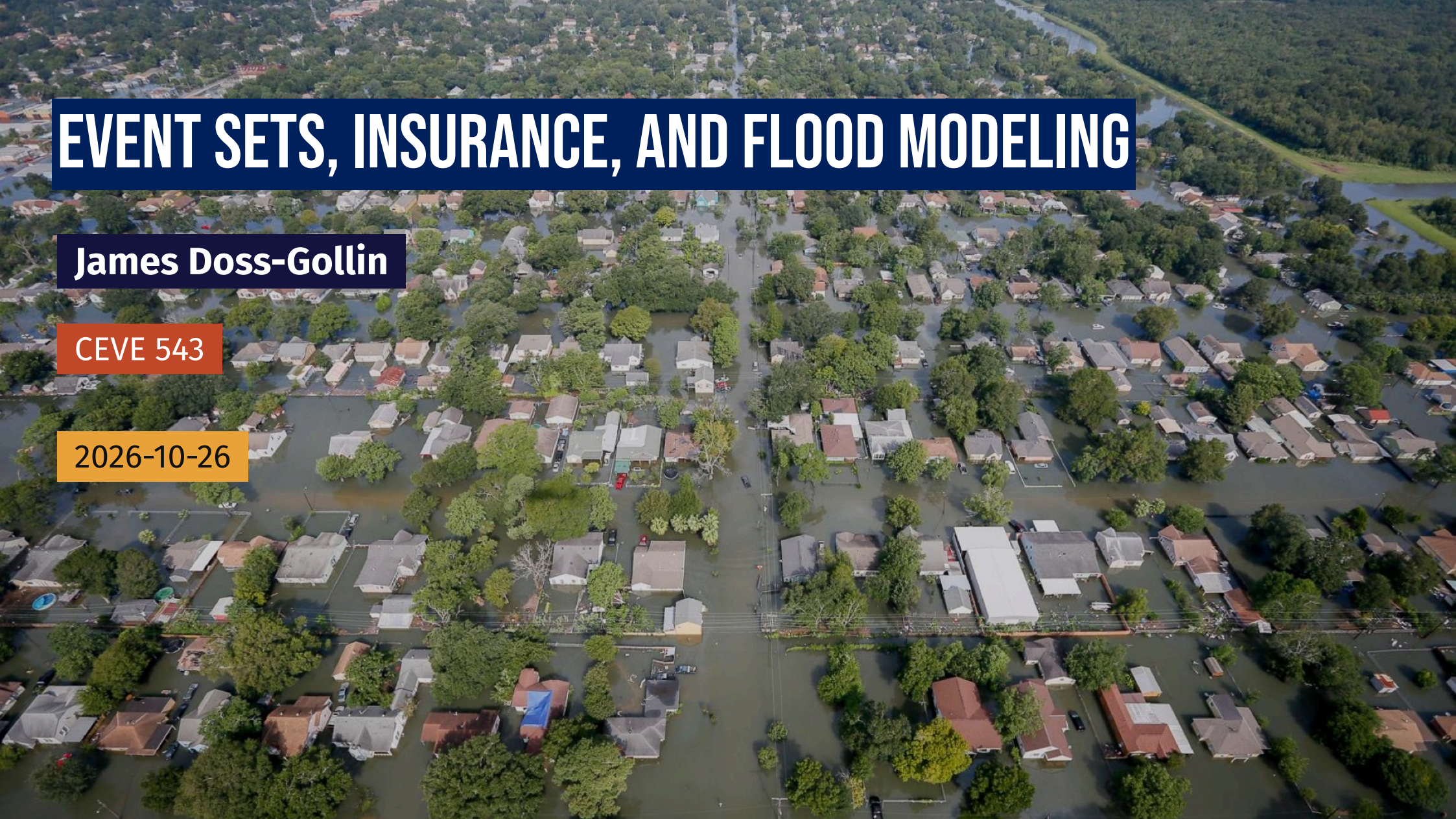
# EVENT SETS, INSURANCE, AND FLOOD MODELING

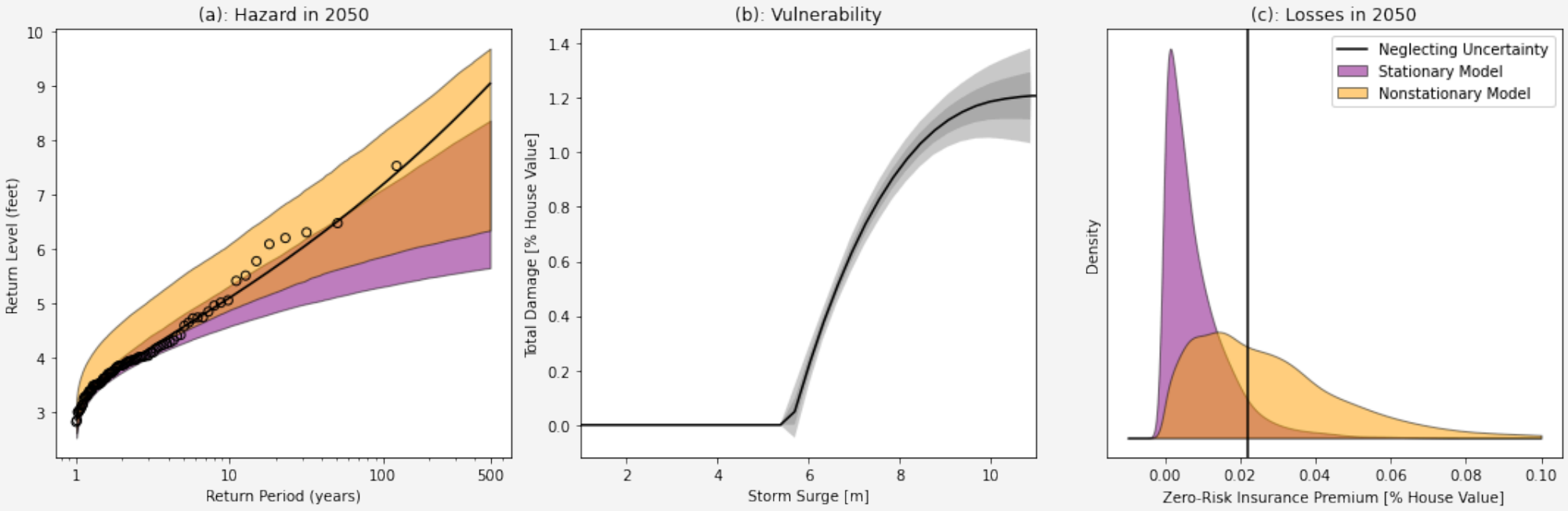
James Doss-Gollin

CEVE 543

2026-10-26



# WHAT IS THE DISTRIBUTION OF FLOOD LOSSES IN A CITY?




Flood hazard, house vulnerability, and the resulting distribution of insurance premiums. Own work.

# COULD WE FIT A GEV OR GPD TO PAST FLOOD LOSSES?

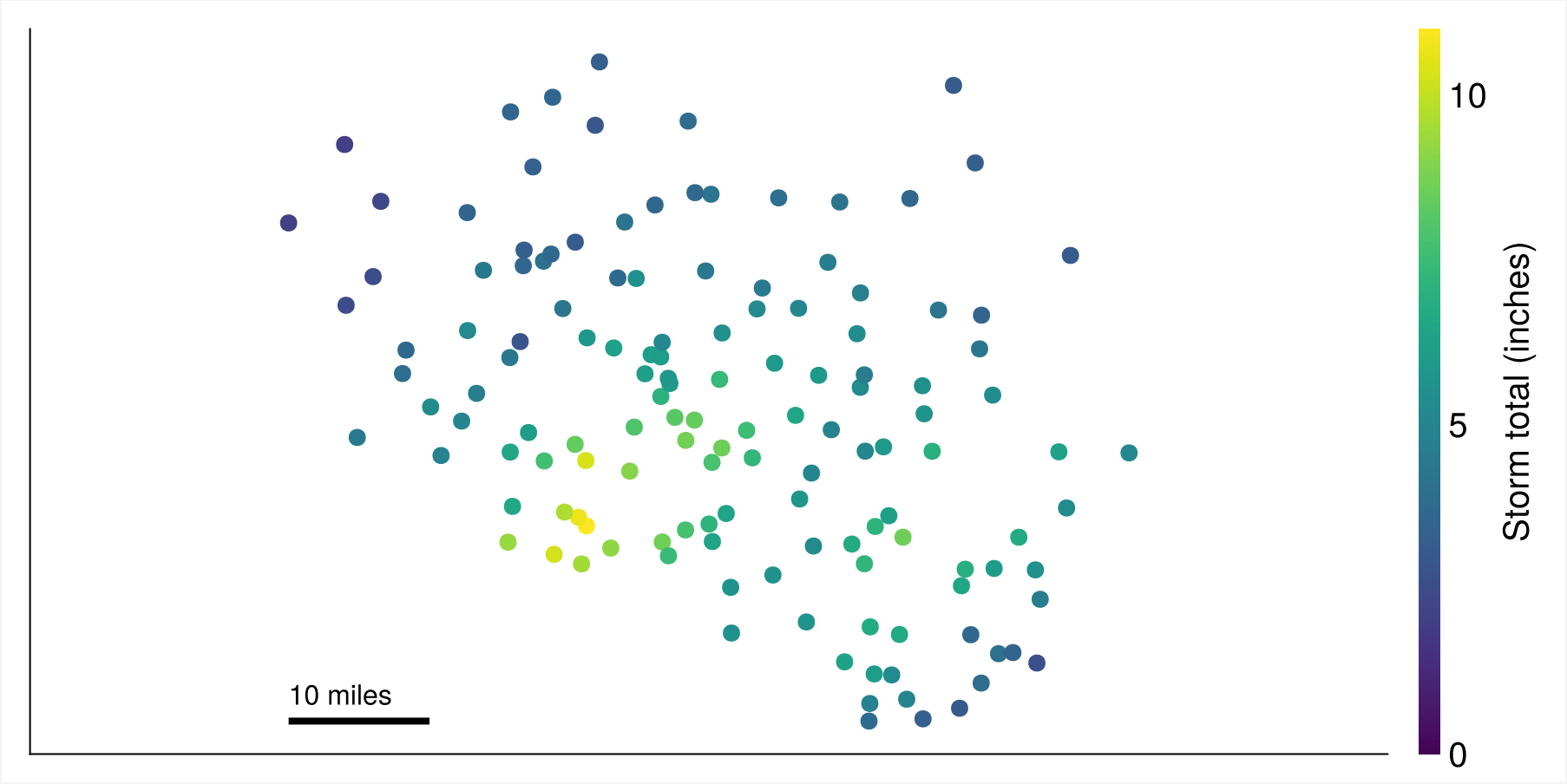


A submerged suburban neighborhood after Hurricane Harvey. Photo by Staff Sergeant Martinez, public domain.

# AN EVENT SET: REALISTIC EVENTS WE CAN COMPUTE DAMAGE FROM

| Event | Rate (per year) | Loss (million dollars) |
|---|---|---|
| 1 | 0.002 | 410 |
| 2 | 0.005 | 160 |
| 3 | 0.010 | 95 |
| 4 | 0.040 | 22 |
| ⋮ | ⋮ | ⋮ |

# MEMORIAL DAY 2015: WHERE DID THE RAIN FALL?




Totals for May 25-26 at Harris County Flood Control District gauges.

# WHERE DID MEMORIAL DAY 2015 FLOOD BUILDINGS?

**i Note**

Figure shown in class: Buildings flooded in 2015 and 2016, many outside FEMA's high-risk zones. Satija et al. (2016).

What was the return period of this flood, and how would it enter average annual loss?

# WHERE COULD HURRICANES HAVE GONE THAT THE RECORD HAS NOT SEEN?

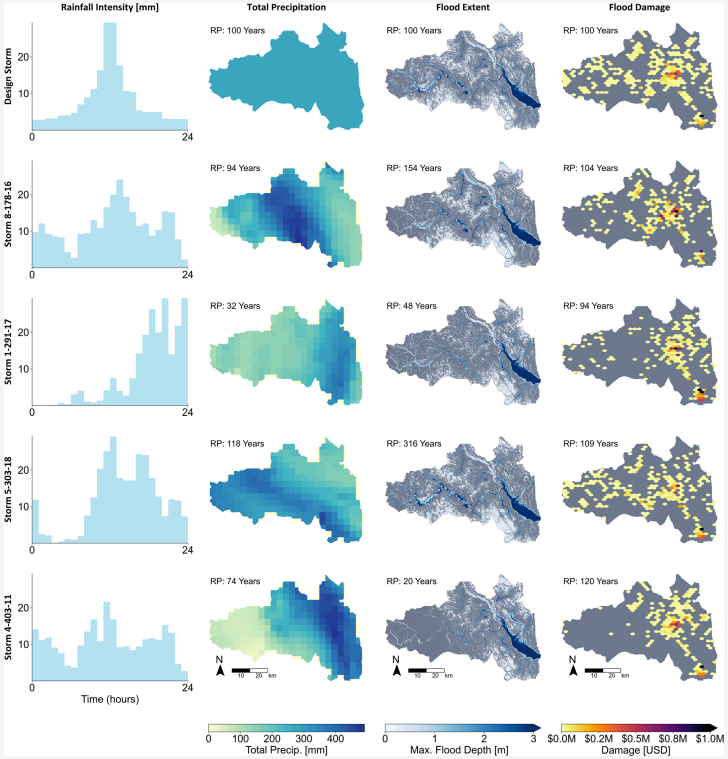
**Note**

Figure shown in class: Historical major hurricane tracks beside simulated tracks from the same coast. O'Rourke & Burdick (2025), p. 5.

# WHERE DO REALISTIC EVENTS COME FROM?

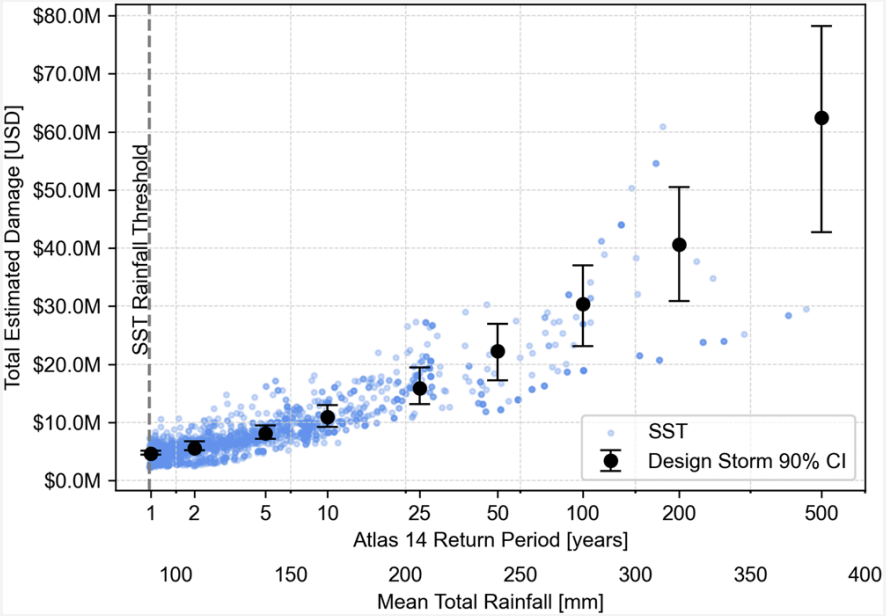
| Storm transposition | Climate models | Weather generators |
| --- | --- | --- |
| This week | Week 11 | Week 13 |

# DOES IT MATTER HOW THE STORMS ARE BUILT?




Four transposed storms with about 100-year damage, and the design storm. Baer et al. (2026), Fig. 4.

# THE SAME RAINFALL?




Damage against basin rainfall, transposition and design storms. Baer et al. (2026), Fig. 2.

Expected annual damage: 14.2 million dollars a year from transposition, 7.4 million more than from design storms (Baer et al., 2026, p. 3).